Shih-Ni Prim
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EDUCATION

PhD North Carolina State University, Statistics — 2022-present
Advisor: Brian J. Reich
Cumulative GPA: 4.0
Expected graduation date: May 2026

MR North Carolina State University, Statistics — Dec 2022
Cumulative GPA: 4.0

COMPUTER SKILLS

Programming: R, Python
Certificate: SAS Certified Specialist: Base Programming Using SAS 9.4 (Issued May 2020)
Tools: High performance computing, Git version control, parallel computing

AWARDS

Glaxo Smith Kline (GSK) Fellowship — 2025

B. B. Bhattacharyya Excellence Award — 2023–2024
North Carolina State University, Department of Statistics

RESEARCH INTERESTS AND EXPERIENCES

Bayesian inference and hierarchical modeling, spatial and spatiotemporal statistics, tensor decomposition, tensor regression, surrogate modeling for computer experiments, Gaussian process, active learning, climate and environmental applications, causal inference

RESEARCH EXPERIENCE

Environmental Protection Agency, Research Triangle Park, NC — August 2024 to present
Oak Ridge Institute for Science and Education (ORISE) Fellow

- Advisors: Ana Rappold and Brian Reich
- Research project: A Multivariate Spectral Model with Adjustment for Unmeasured Confounders for Multiple Outcomes and Multiple Exposures
- Proposed a spectral, fully Bayesian model to adjust for spatial confounding and ensure causal interpretation of effects
- Constructed fully Bayesian MCMC algorithms in R with different priors and capacity (multivariate vs univariate outcomes) to compare model performance
- Used tensor structure so that coefficients can vary by spatial scales, exposures, and outcomes
- Implemented canonical polyadic tensor decomposition to reduce dimension
- Performed extensive simulation studies
- Visualized Geographic Information System data using R (sf, tidyverse, and ggplot2)
- Paper uploaded to arXiv and under review at Journal of the American Statistical Association

Collaborator of Lawrence Livermore National Laboratory August 2024 to present

- Advisors: Kevin Quinlan and Annie Booth
- Research project: Active Learning with Deep Gaussian Processes for Aero Database Construction
- Examined predictive accuracy of Deep Gaussian Processes versus ordinary Gaussian Processes as surrogate models for aerodynamic data
- Devised active learning algorithms to efficiently construct aero databases in Python (scikit-learn, numpy, dgpsi)
- Created visualization of the sequential sampling process and results in Python (matplotlib)
- Presented preliminary results at conference
- Currently drafting a manuscript

Lawrence Livermore National Laboratory, Livermore, CA May to August 2024
Data Science Summer Institute Intern

- Conducted literature review on surrogate modeling, contour estimation, and active learning
- Presented at end-of-summer presentation for interns

North Carolina State University, Raleigh, NC August 2022 to May 2024
Research Trainee at Duke Clinical Research Institute

- Helped with developing and evaluating analytic methods based on WIN statistics
- Shadowed meetings of clinical research and received training on ethics of research

North Carolina State University, Raleigh, NC Summer 2022, July 2023 to May 2024
Research Assistant

- Principal Investigators: Natalie Nelson and Brian Reich
- Used statistical and machine learning models to analyze spatiotemporal data to examine inference about harmful algal blooms
- Data cleaning and preprocessing, data analysis, statistical modeling, manuscript drafting

Wake Forest Baptist Health, Winston-Salem, NC 2021 to 2022
Volunteer Research Statistician / Intern

- Conducted Kaplan Meier Survival Analysis and Cox Proportional Hazard Model on retrospective data of patients with glioblastoma
- Created tables and plots and help with writing and editing for publication
- Helped collect MRI images and clinical data for studies that utilize machine learning for classification

PUBLICATIONS

Prim, S.-N., Guan, Y., Yang, S., Rappold, A. G., Hill, K. L., Tsai, W.-L., Keeler, C. and Reich, B. J. (2025) “A Spectral Confounder Adjustment for Spatial Regression with Multiple Exposures and Outcomes,” arXiv:2506.09325.

Nelson N, **Prim S.-N.**, Saia S, Grieger K, Huseth A (authors vary by chapter) (2025), A Machine Learning Primer for Natural Resources Management, accessed online via go.ncsu.edu/mlprimer.

Chan MD, **Prim S.-N.**, Cramer C, Ruiz J. (2023), “Nonrandomized trials in clinical oncology,” in Handbook for Designing and Conducting Clinical and Translational Research: Translational Radiation Oncology, edited by Adam E. M. Eltorai, Jeffrey A. Bakal, Daniel W. Kim, David E. Wazer, Academic Press, pp. 313–284.

Helis CA, **Prim S.-N.**, Cramer CK, Strowd R, Lesser GJ, White JJ, Tatter SB, Laxton AW, Whitlow C, Lo H, Debinski W, Ververs JD, Black PJ, Chan MD (2022), "Clinical outcomes of dose-escalated re-irradiation in patients with recurrent high-grade glioma," Neuro-Oncology Practice, 9, 390–401.

CONFERENCE PRESENTATIONS

Prim S.-N., Guan, Y., Yang, S., Rappold, A. G., Hill, K. L., Tsai, W.-L., Keeler, C. and Reich, B. J. "A Spectral Confounder Adjustment for Spatial Regression with Multiple Exposures." International Workshop on Applied Probability, June 10, 2025.

Prim S.-N., Quinlan K, Booth A. "Active Learning with Deep Gaussian Processes for Aero Database Construction." International Conference on Advances in Interdisciplinary Statistics and Combinatorics, October 13, 2024.

GRANTS AND FELLOWSHIPS

ORISE Fellowship — 2024 to present
Environmental Protection Agency

T32 Training Grant — 2022 to 2024
Duke Clinical Research Institute

LANGUAGES

English, Mandarin, German, French, Italian

REFERENCES

Brian Reich, Gertrude M Cox Distinguished Professor of Statistics
Department of Statistics, North Carolina State University
Phone: 919-513-7686
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Yawen Guan, Assistant Professor
Department of Statistics, Colorado State University
Email: yawen.guan@colostate.edu

Natalie Nelson, Associate Professor
Department of Biological and Agricultural Engineering, North Carolina State University
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Justin Post, Associate Teaching Professor, Director of Statistics Online Education
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